(Please write your Enrollment Number) Enrollment No. 01601172021

MID-TERM EXAMINATION
(Course Name : B. Tech CSE-AI) (Semester : VI)
(March, 2024) OFF LINE mode

Subject Code: BAI - 306	**Subject: Digital Image Processing**
Time : 1 ½Hours	**Maximum Marks : 30**
Note: Q. 1 is compulsory.	

Q1		**(2.5*4)**	
	(a) What is the difference between analog image and digital image? What are the applications of digital image processing?	UNIT-1	
	(b) Distinguish between sampling and quantization.	UNIT-1	
	(c) Why is median filter is better than mean filter?	UNIT-2	
	(d) Define homomorphic filtering with equations.	UNIT-2	

Q2 **(Attempt any Two Parts) UNIT-1** **(5,5)**

(a) Explain fundamental steps of Digital Image processing.

(b) The following image has size 5 x 5 and its gray levels varies from 0 to 7. Perform histogram equalization and obtain the final image.

5	5	5	5	5
3	5	7	5	3
3	7	7	7	3
3	5	7	5	3
5	5	5	5	5

(c) Consider the following image segment. *Compute Dm, D4 and D8 distances between pixels 'p' and 'q' for V ={0,1} where V is the set of grey values used to define connectivity. Repeat for V = {1,2}*

3	1	2	1
0	2	0	2
1	2	1	1
1	0	1	2

Q3 **(Attempt any Two Parts) UNIT-2** **(5,5)**

(a) Compute 2D DCT of the gray scale image is given by

4	2	9	60
7	10	5	77
88	66	44	3
1	1	1	1

(b) Perform contrast stretching on the given image

7	7	8	8
6	7	6	9
8	7	6	9
6	6	7	7

$r_1 = 6$ $s_1 = 3$
$r_2 = 8$ $s_2 = 7$

(C) Find the deblur filter using Wiener filter approach with $\sigma^2_x = 100$ and $\sigma^2_y = 200$. A blur filter is given by h(m,n):

0	0.05	0.05	0
0.15	0.1	0.1	0.15
0	0.1	0.1	0
0	0.1	0.1	0